

Introduction to Artificial Intelligence– CII2M3

11-Nearest Neighbor



<NAMA DOSEN> - <KODE DOSEN> - <KELAS>

(per sesi 50 menit, kemudian break +/- 10 menit)



MINGGU KE-	ID CLO	DESKRIPSI SUB CLO	INDIKATOR KETERCAPAIAN CLO	BENTUK ASSESMENT	MATERI	METODE
12	CLO 3	•Mahasiswa dapat menjelaskan cara kerja teknik learning. • Mahasiswa dapat menganalisis dan mendesain sebuah penyelesaian masalah menggunakan teknik learning. •Mahasiswa mampu mengimplementasikan sebuah teknik learning untuk menyelesaikan masalah yang diberikan.	•Ketepatan dalam menjelaskan konsep k-nearest neighbor (kNN). •Ketepatan analisis dan desain Jarak Atribut Numerik, Non Numerik, Campuran. •Efektivitas dan efisiensi dalam implementasi Algoritma kNN.	Kerja Lab	•k-nearest neighbor (kNN) •Jarak Atribut Numerik •Jarak Atribut Non Numerik •Jarak Atribut Campuran •Algoritma kNN •Optimasi k •Perbaikan kNN Pustaka: utama.	• Ceramah Kuliah. • Diskusi. • Kerja Lab 1

Nearest Neighbor



Perhatikan ketiga kura-kura (atau penyu) diatas. Kemudian sebutkan apa saja yang membuat ketiganya merupakan termasuk kategori kura-kura/penyu walaupun tidak memiliki bentuk fisik yang serupa?

Setelah menemukan kesamaannya, mari dilanjutkan mempelajari mengenai "NEAREST NEIGHBOR"

Enormous Attributes

- Decision tree is good for **limited and discrete attributes**
- In case of classification with numerous attributes,
the tree resulted might be too complex and prone to overfit
- For example: in image classification

Character Recognition

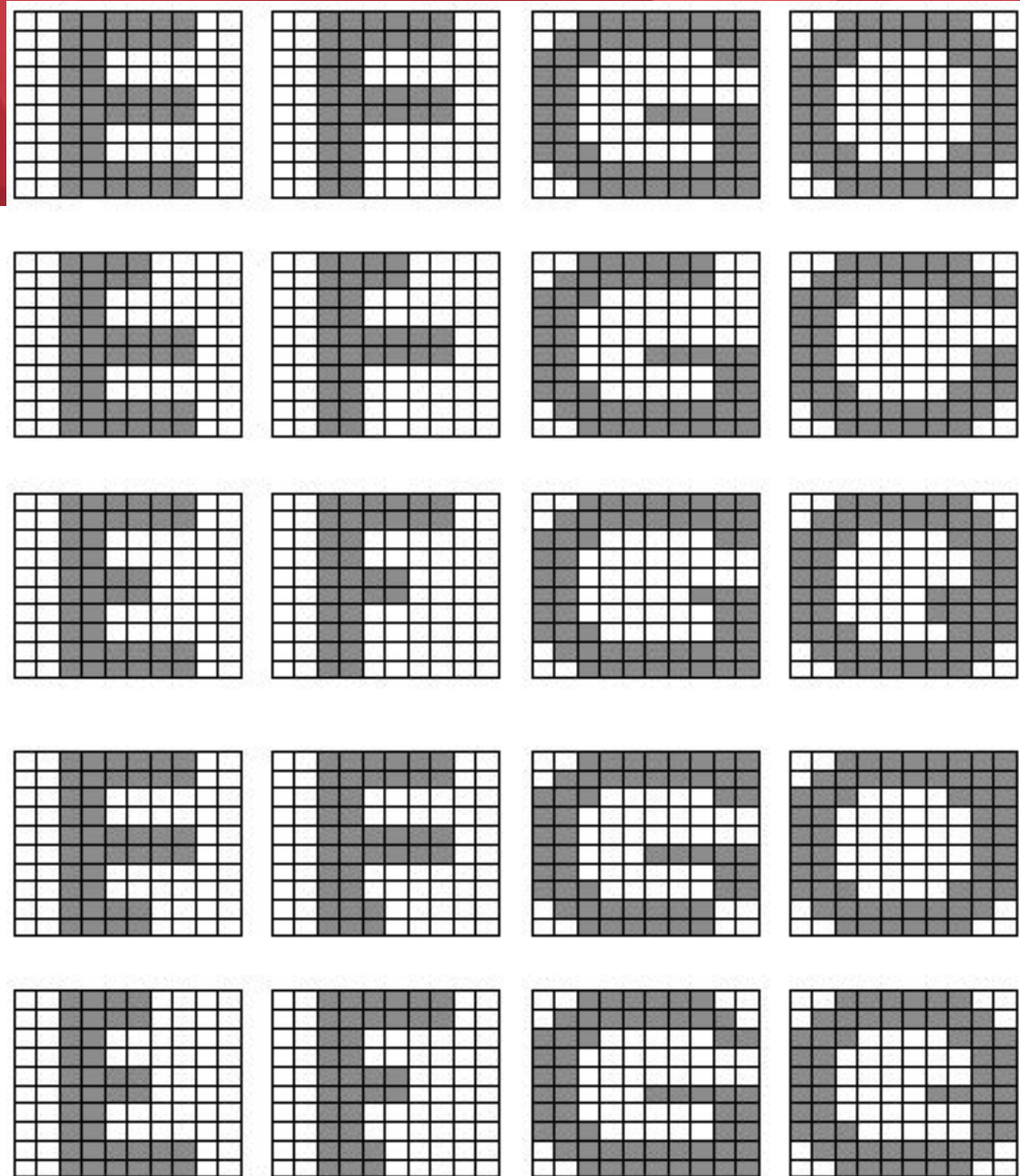
Perhatikan 4 huruf E, F, G, O

Diketahui:

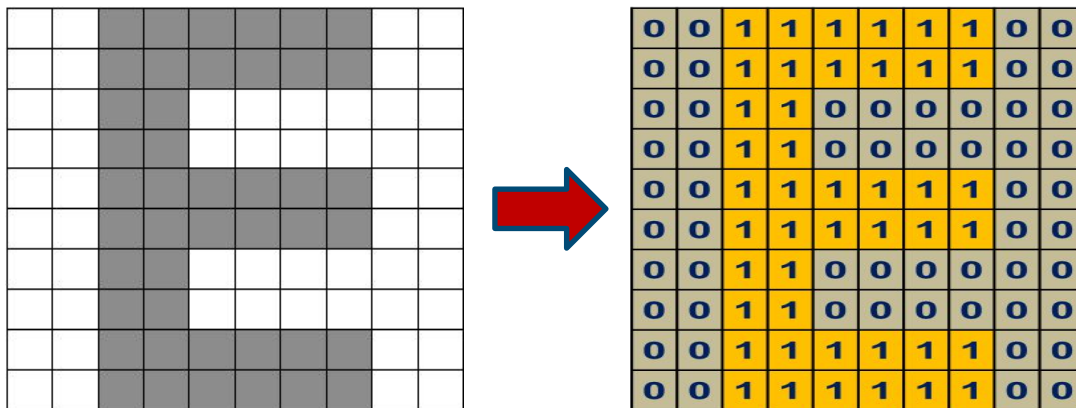
Terdapat 4 class, yakni E, F, G, O

Jumlah data = 20 (dari 4 kelas X 5 baris data)

Banyak atribut/fitur 1 data yaitu
10x10 pixel



Character Recognition



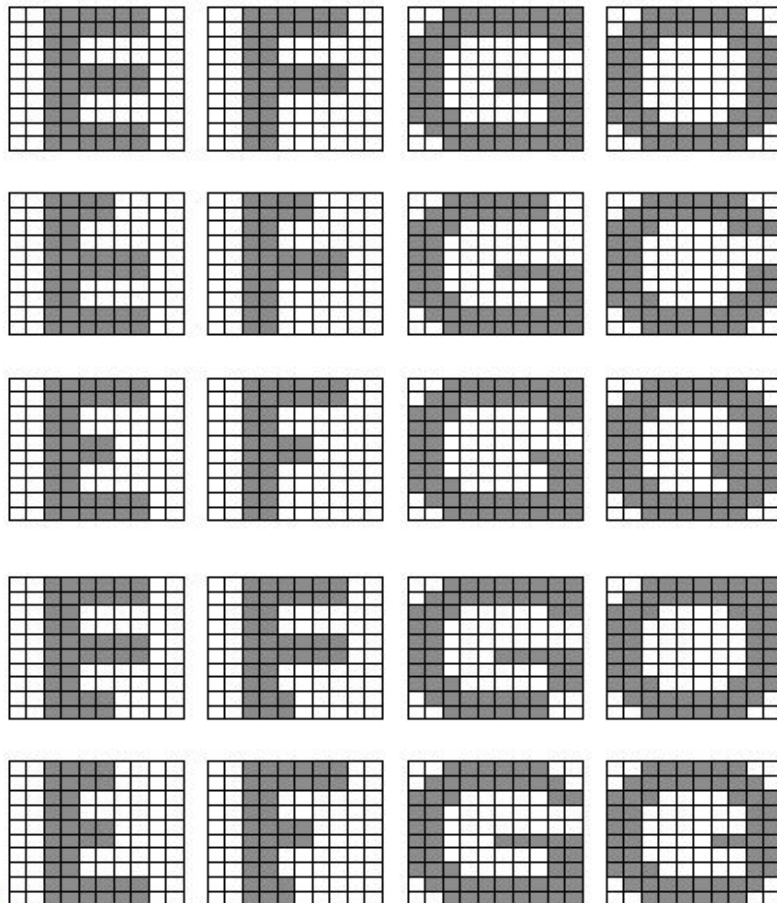
To ease the computation, change data into 1D array





20 data
100 dimensional
4 class

Character Recognition Dataset



	Pix 1	Pix 2	Pix 3	Pix 4	Pix 5	...	Pix 100
E1	0	0	1	1	1	...	0
F1	0	0	1	1	1	...	0
G1	0	0	1	1	1	...	0
O1	0	0	1	1	1	...	0
..							
O5	0	0	1	1	1	...	0

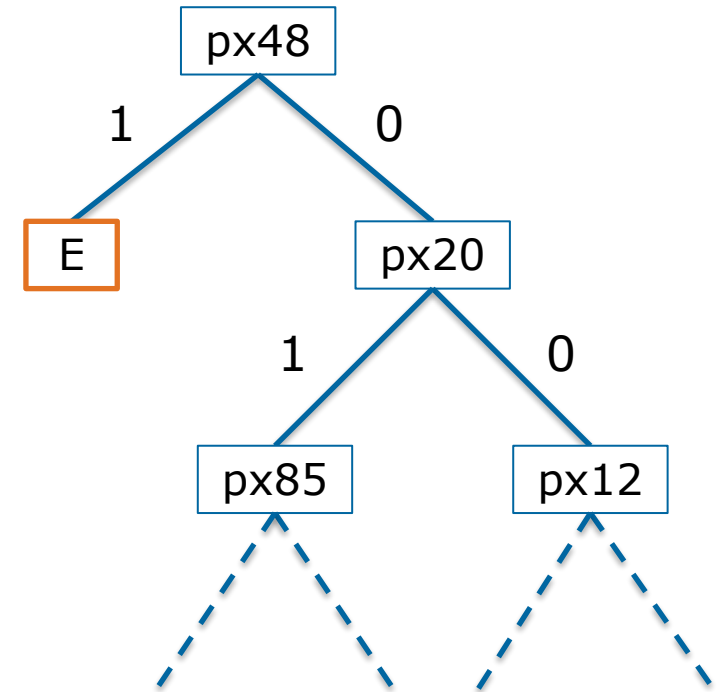


Character Recognition – ID3

- Let's say we try to build character classifier based on pixel input using ID3
- What will happen?

Character Recognition – ID3

- The tree will be too long
- A complete ID3 will result a 100-level-depth tree (100-dimensional input)
- Not efficient





Nearest Neighbor

Nearest Neighbor

- Simple algorithm that **stores all available cases** and classifies new cases based on a similarity measure
- **Non-parametric** techniques
- Other names:
 - **Memory-Based Reasoning**
 - Example-Based Reasoning
 - Instance-Based Learning
 - Case-Based Reasoning
 - **Lazy Learning**



Train



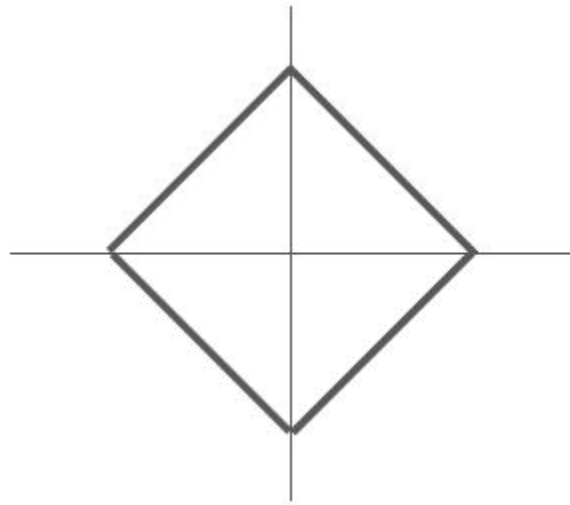
Nearest Neighbor

- Training Process
 - Memorize all training data and labels
- Testing Process
 - Use **L1** or **L2** (or other distance choices) distance to each and every training data, return label of the closest one
 - L1 : Manhattan Distance
 - L2 : Euclidean Distance

Distance Metric

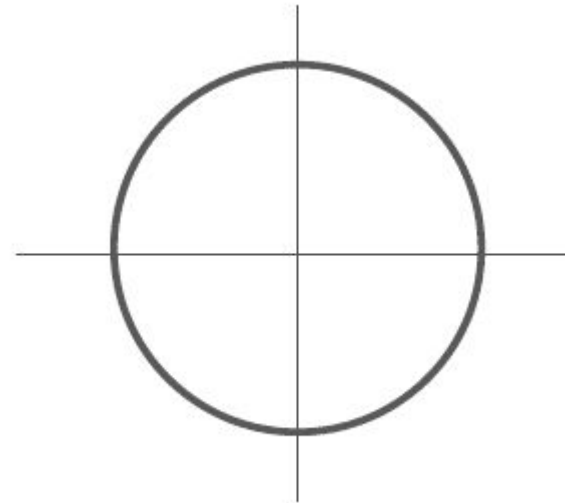
L1 (Manhattan) distance

$$d_1(x_1, x_2) = \sum_p |x_{1p} - x_{2p}|$$



L2 (Euclidean) distance

$$d_1(x_1, x_2) = \sqrt{\sum_p (x_{1p} - x_{2p})^2}$$



Nearest Neighbor

L1 distance: $d_1(x_1, y_2) = \sum_p |x_1^p - y_2^p|$

test data (x)				train data (y)				value differences			
1	1	1	1	1	1	1	0	0	0	0	1
1	0	0	0	1	0	0	1	0	0	0	1
1	1	1	0	1	0	0	1	0	1	1	1
1	0	0	0	1	1	1	0	0	1	1	0

- = add **7**

Nearest Neighbor

L2 distance : $d_1(x_1, y_2) = \sqrt{\sum_p (x_1^p - y_2^p)^2}$

test data (x)			
1	1	1	1
1	0	0	0
1	1	1	0
1	0	0	0

-

train data (y)			
1	1	1	0
1	0	0	1
1	0	0	1
1	1	1	0

=

value differences			
0	0	0	1
0	0	0	1
0	1	1	1
0	1	1	0

→

sqrt(add) $\sqrt{7} = 2.645$

Nearest Neighbor

L1 distance : $d_1(x_1, y_2) = \sum_p |x_1^p - y_2^p|$

test data (x)				train data (y)				value differences			
1	1	1	1	1	1	1	0	0	0	0	1
1	0	0	0	1	0	0	1	0	0	0	1
1	1	1	0	1	1	1	1	0	0	0	1
1	0	0	0	1	0	0	0	0	0	0	0

- = add **3**

Nearest Neighbor Algorithm

- For each test data A ,
 - loop through all training data, calculate the distance
 - Choose the closest training data point X
 - Set the label of X as the label of data test A
- Nearest neighbor intuition:
 - Since A is similar to X (the attributes are close), then A and X must be in the same class

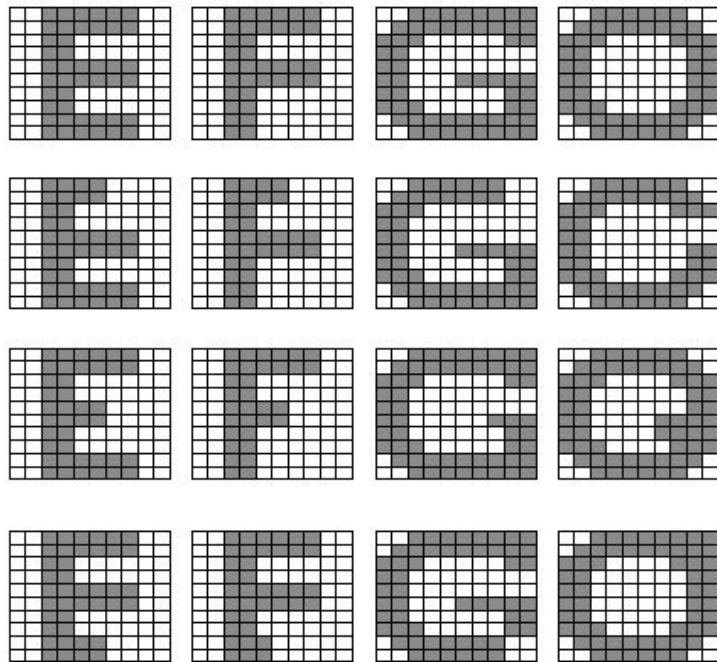


Character Recognition with Nearest Neighbor

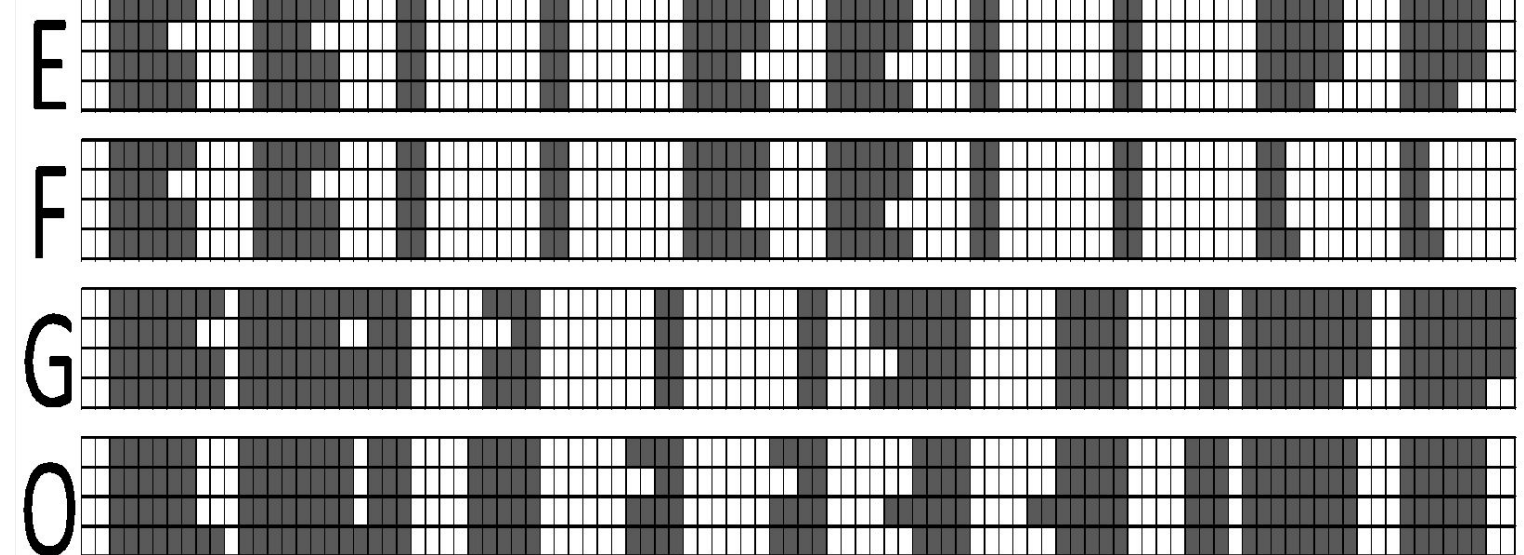
Training and Testing

Nearest Neighbor Learning Process

- Let's say we use 16 data as training data:

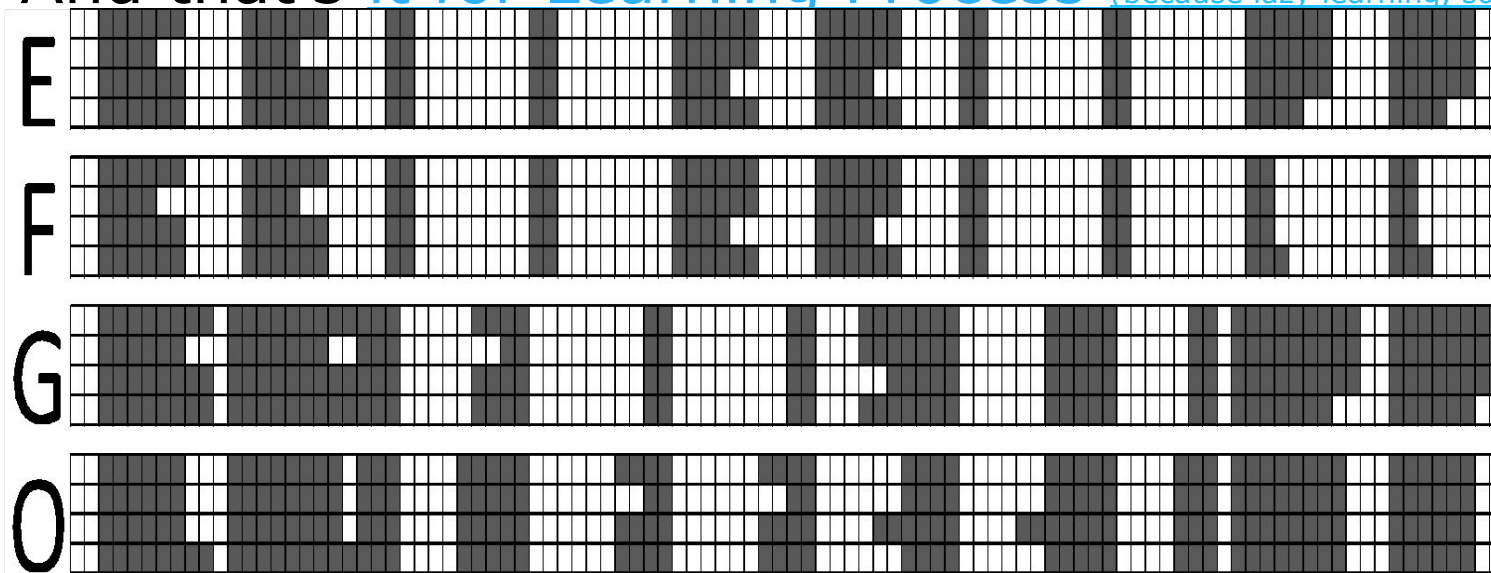


- Reshape into vectors



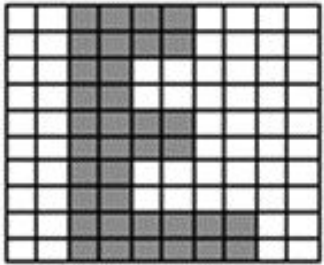
Nearest Neighbor Learning Process

- Save to database,
- And that's [it for Learning Process](#) (because lazy learning, so it's just memorizing all training data and labels)



Nearest Neighbor Testing Process

- Now we want to classify **new data** using Nearest Neighbor

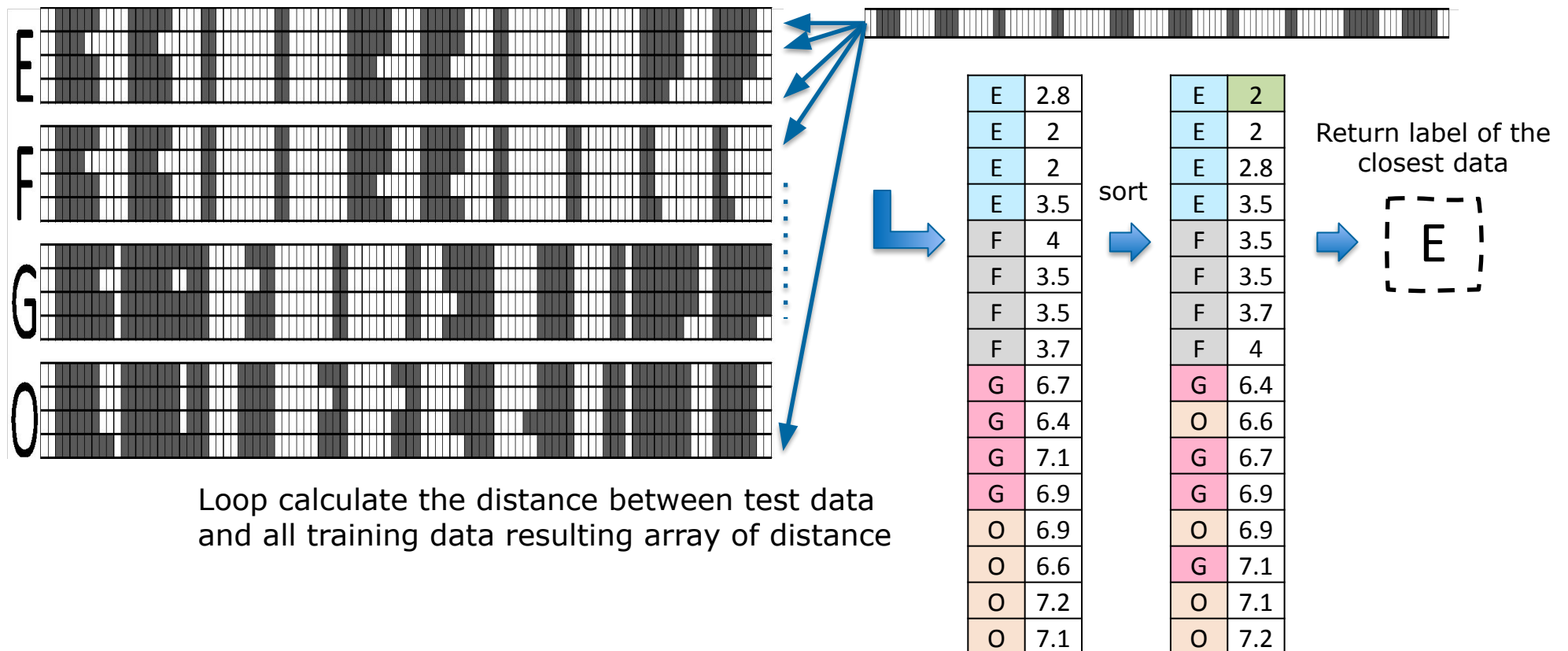


First, we change new data into 1 dimensional



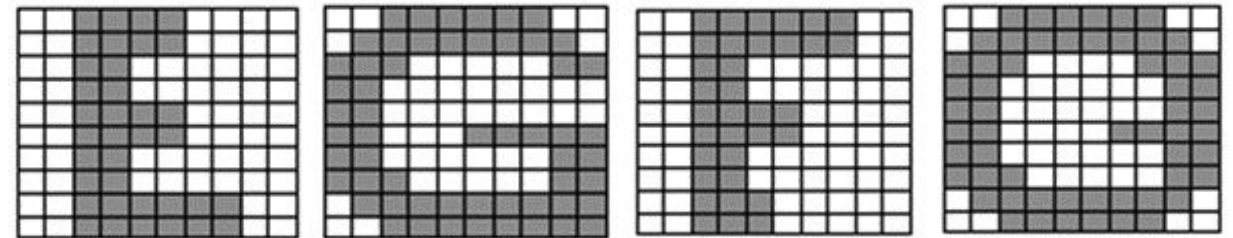
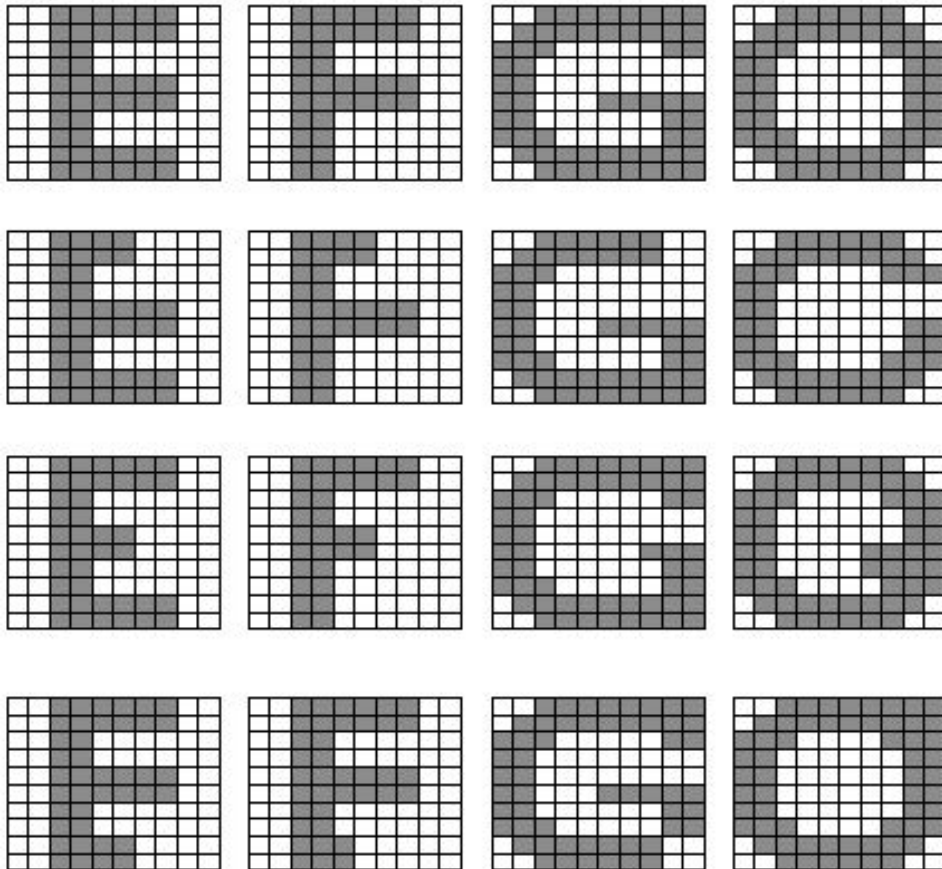
Then calculate distance (use L1 or L2) to the 16 data training

Nearest Neighbor Testing Process





Nearest Neighbor Testing Process



E				F				G				O			
2.8	2	2	3.5	4	3.5	3.5	3.7	6.7	6.4	7.1	6.9	6.9	6.6	7.2	7.1
3.2	3.7	2.4	2.4	2.4	3.2	1.4	2	6.9	6.6	7.3	7	7.1	6.8	7.3	7.3
6.6	6.9	6.6	6.9	7.2	7.5	7.2	7.1	1.7	2.6	1.7	2.6	3.2	2.4	3.2	3.6
6.8	7.1	6.8	7.1	7.3	7.6	7.3	7.2	2.2	3	3.3	3	1.4	2.4	1.4	2.2



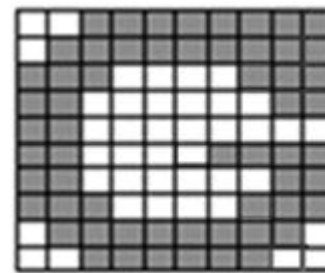
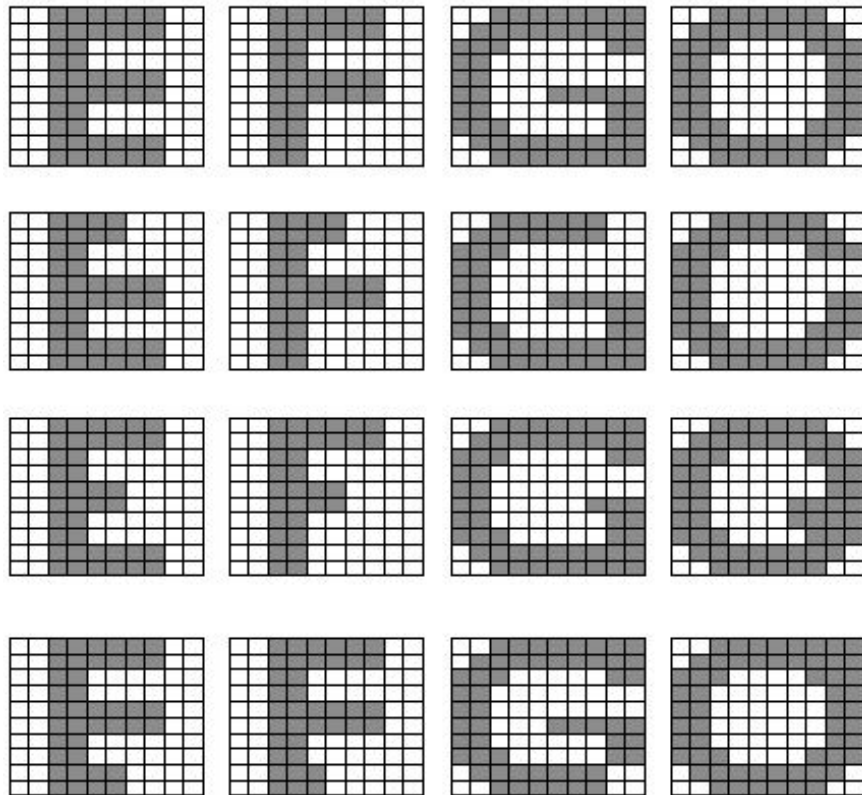
Nearest Neighbor Problem



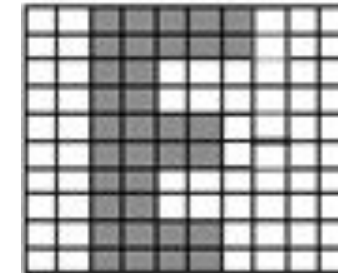
Nearest Neighbor

- Training Speed?
 - $O(1)$
 - No training, just save all training data
- Testing Speed?
 - $O(N)$
 - Linearly slower according to the size of training data

Character Recognition using Nearest Neighbor



G? O?



E? F?

E	E	E	E	F	F	F	F	G	G	G	G	O	O	O	O
6.9	7.1	6.9	7.1	7.4	7.7	7.4	7.3	2.4	3.2	2.4	2	2.6	2.6	2.6	2
3.2	3.2	2.4	2.4	3.2	3.2	2.4	2.8	6.9	6.6	7.3	7	7.1	6.8	7.3	7.3

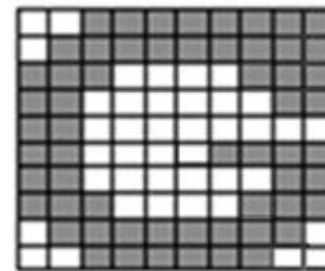
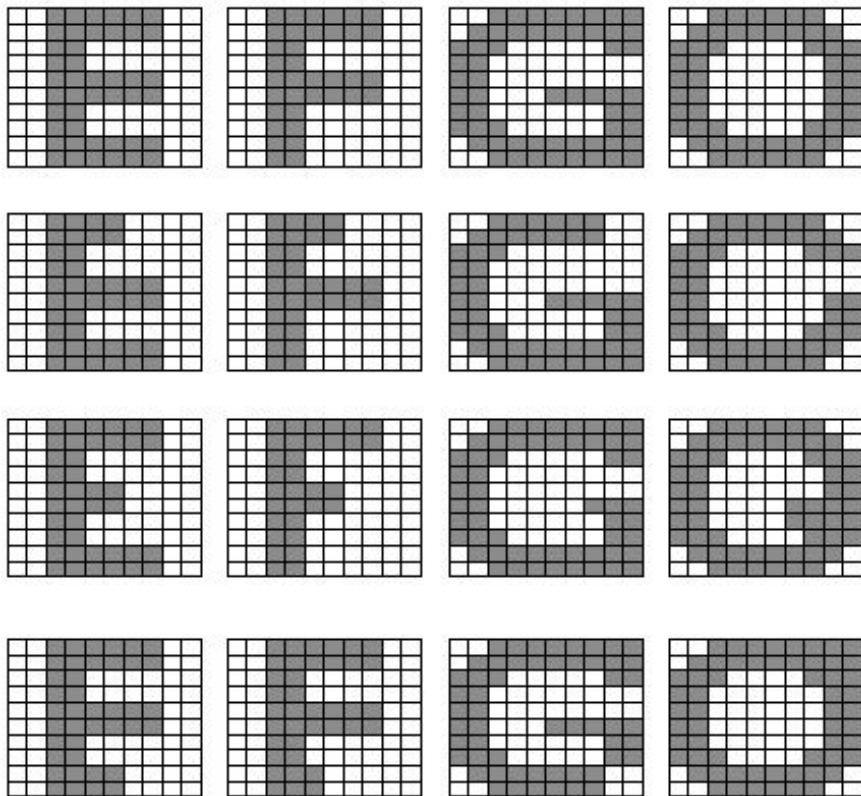


k-Nearest Neighbor

K-Nearest Neighbor

- Use **many neighbors** to determine the class
 - Choose the k-closest data point
 - Majority vote for the label
- K-NN can be used for **Classification and Regression**
 - For classification select the most frequent neighbor.
 - For regression calculate the average of K neighbors

Character Recognition using k-NN



$K = 3$

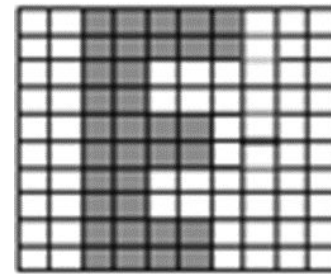
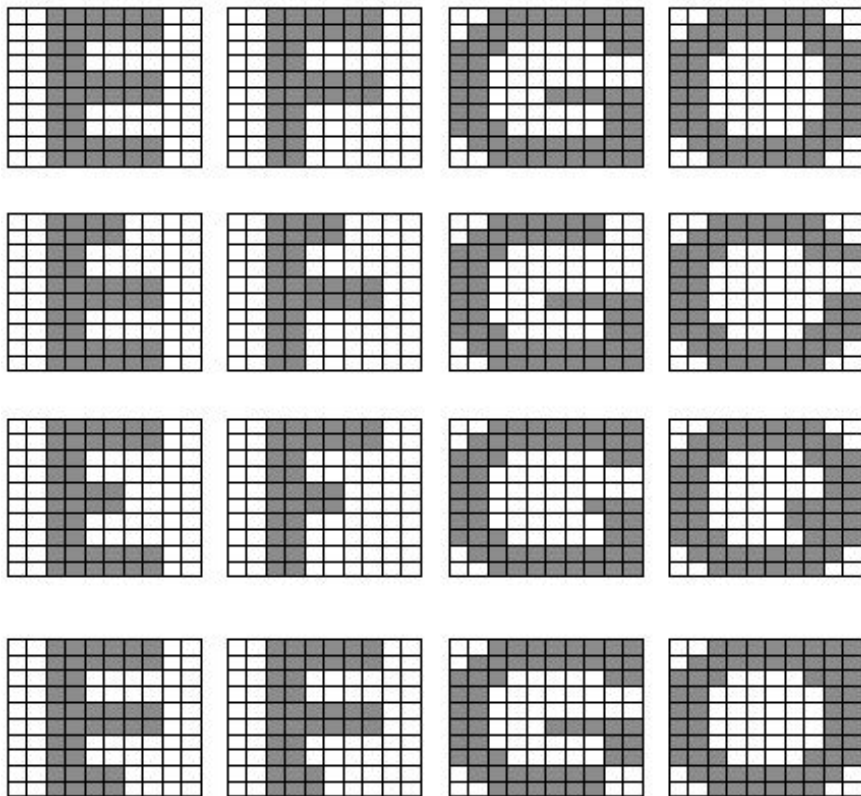
E	E	E	E	F	F	F	F	G	G	G	G	O	O	O	O
6.9	7.1	6.9	7.1	7.4	7.7	7.4	7.3	2.4	3.2	2.4	2	2.6	2.6	2.6	2

sort

G	O	G	G	O	O	O	G	E	E	E	E	F	F	F	F
2	2	2.4	2.4	2.6	2.6	2.6	3.2	6.9	6.9	7.1	7.1	7.3	7.4	7.4	7.7

2G, 1O = G

Character Recognition using k-NN



K = 3

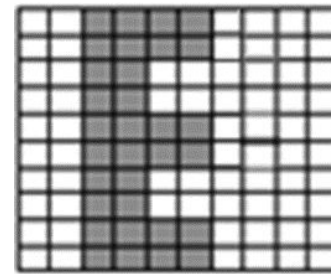
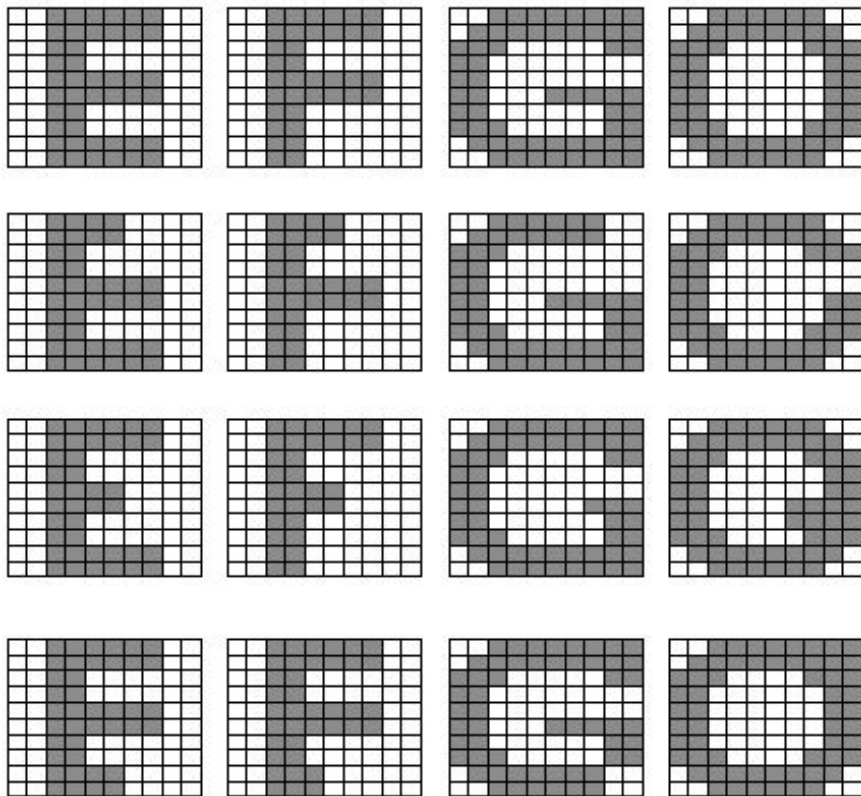
E	E	E	E	F	F	F	F	G	G	G	G	O	O	O	O
3.2	3.2	2.4	2.4	3.2	3.2	2.4	2.8	6.9	6.6	7.3	7	7.1	6.8	7.3	7.3

sort

E	E	F	F	E	E	F	F	G	O	G	G	O	G	O	O
2.4	2.4	2.4	2.8	3.2	3.2	3.2	3.2	6.6	6.8	6.9	7	7.1	7.3	7.3	7.3

2E, 1F = E

Character Recognition using k-NN

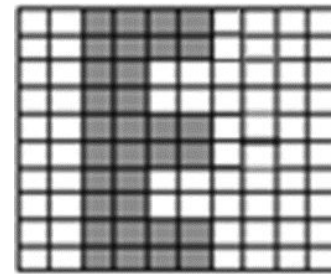
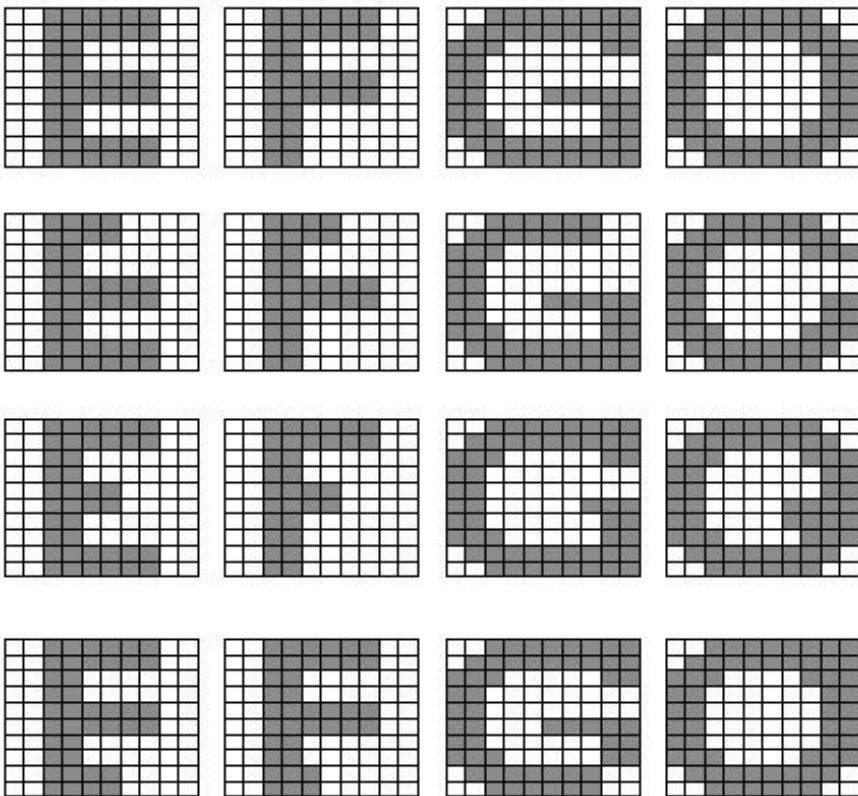


$K = 3$

E	E	E	E	F	F	F	F	G	G	G	G	O	O	O	O
3.5	2.8	2.8	2.8	3.5	2.8	2.8	3.2	7	6.7	7.4	7.1	7.2	6.9	7.5	7.4

E ?? F ??

Character Recognition using k-NN



$K = 3$

E ?? F ??

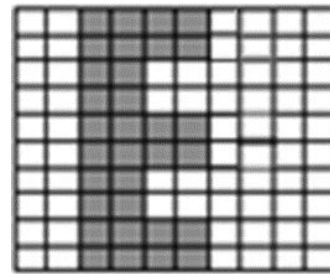
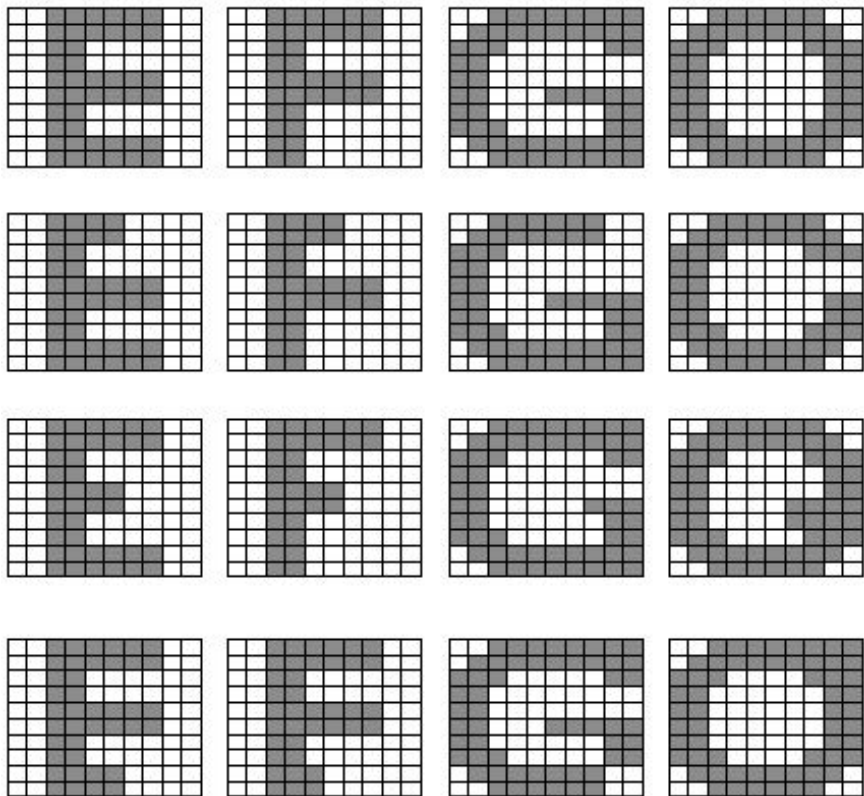
sort

E	F	E	E	F	F	E	F	G	O	G	G	O	G	O	O
2.8	2.8	2.8	2.8	2.8	3.2	3.5	3.5	6.7	6.9	7	7.1	7.2	7.4	7.4	7.5

what will happened if the sorting result is

E	F	F	E	E	F	E	F	G	O	G	G	O	G	O	O
2.8	2.8	2.8	2.8	2.8	3.2	3.5	3.5	6.7	6.9	7	7.1	7.2	7.4	7.4	7.5

Character Recognition using k-NN



K = 3 Change □ K = 5

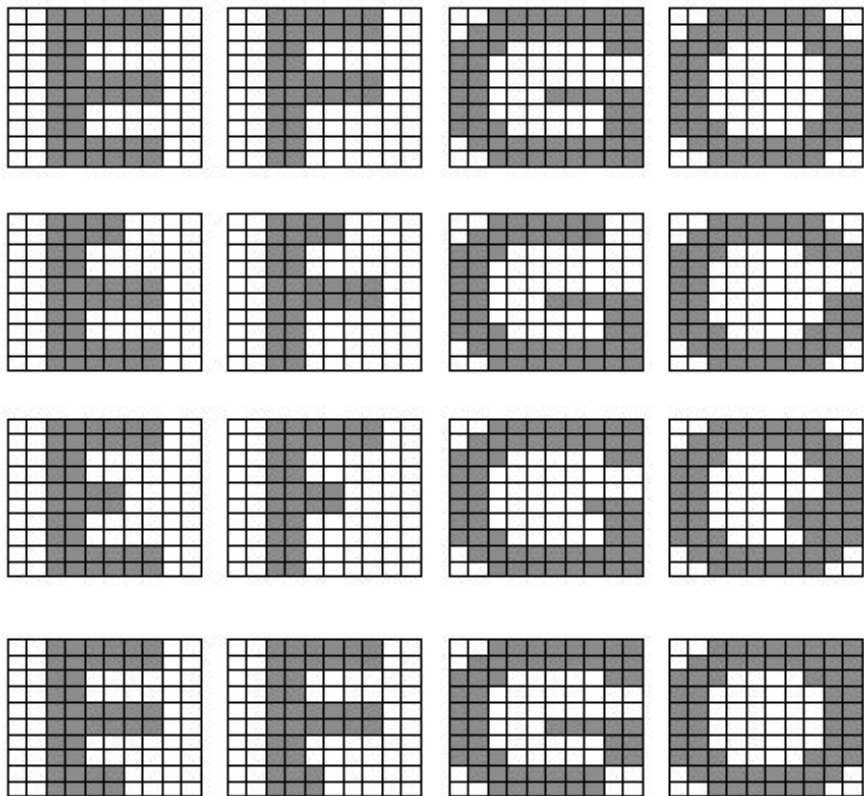
sort

E	F	E	E	F	F	E	F	G	O	G	G	O	G	O	O
2.8	2.8	2.8	2.8	2.8	3.2	3.5	3.5	6.7	6.9	7	7.1	7.2	7.4	7.4	7.5

3E, 2F = E



Character Recognition using k-NN



K = 3?

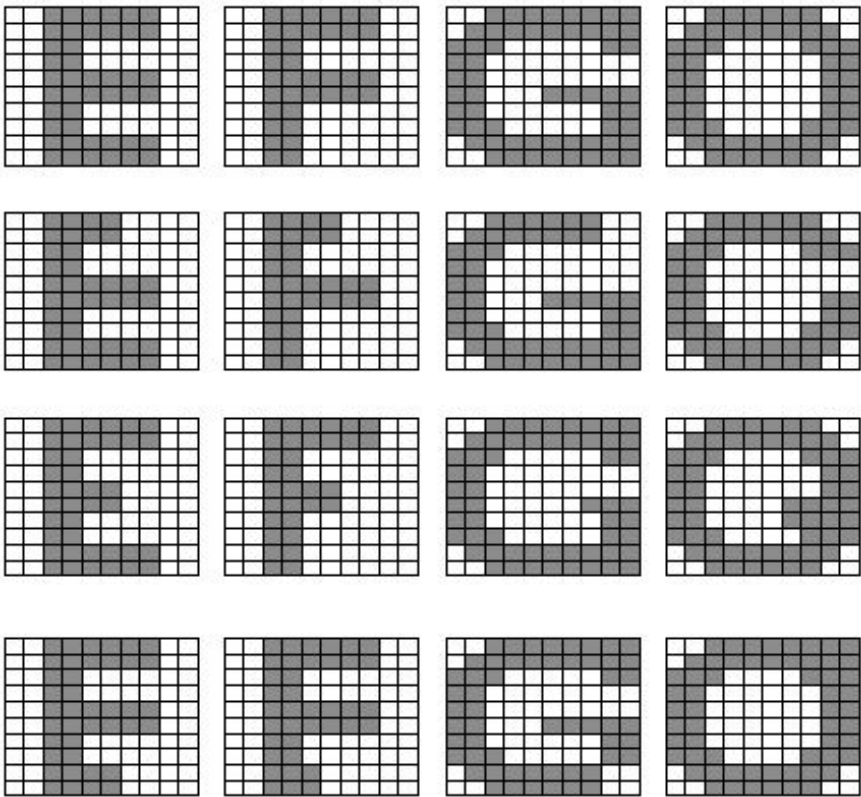
K = 5?

K = 7?

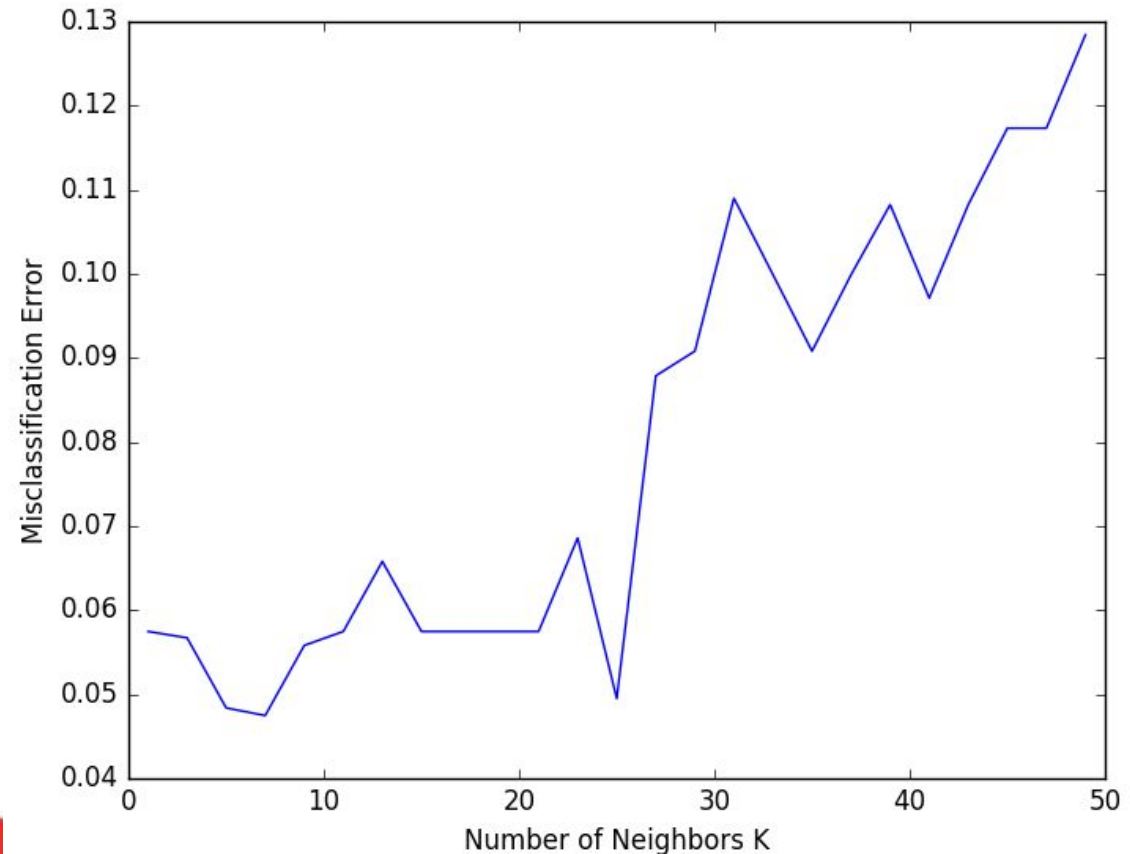
K = 10?

Which one is better?

Character Recognition using k-NN



Observe and determine k using validation set





k-Nearest Neighbors - Hyperparameters

- What is the best **distance** to use?
- What is the best value of **k** to use?
- Choices about the algorithm that we set rather than learn
- Very problem-dependent.
- Must try them all out and see what works best.



K-Fold Cross Validation

Hyperparameter Observation on Trainset?

- We want to observe $k=\{1, 2, 3, 4, 5\}$
- What would happen if we check the performance on train set?
- Best performance will ALWAYS be $k=1$
- Myth: k value should be an odd-number

Note on k -NN Hyperparameter Observation

K-Fold Val: Hyperparameter Observation

- For each iteration
 - Try all hyperparameters
 - Record the validation acc
 - At the end of iteration, average the validation acc
 - Choose best hyperparameter
 - Use the chosen hyperparameter to test new data using ALL data (train+val)
- | | kNN validation | | | | |
|-----------|----------------|---|---|---|---|
| Iteration | 1 | 2 | 3 | 4 | 5 |

	kNN validation				
Iteration	1	2	3	4	5

Question?

Next Agenda: Naïve Bayes



Video Kerja Lab

- Naïve Bayes: [AI Kerja Lab Tutorial Naive Bayes – YouTube](#)
- KNN: [AI Kerja Lab Tutorial KNN - YouTube](#)





THANK YOU